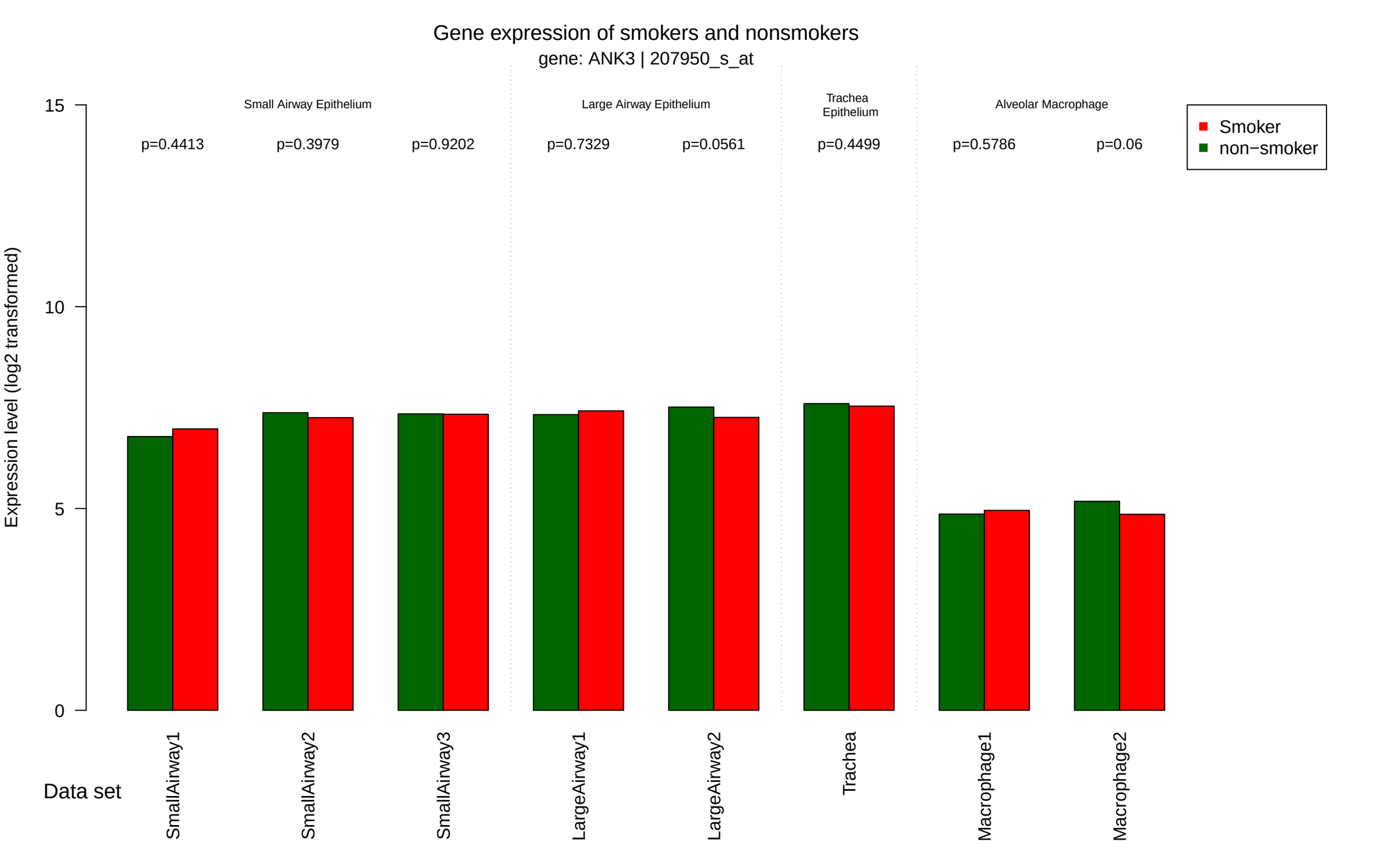
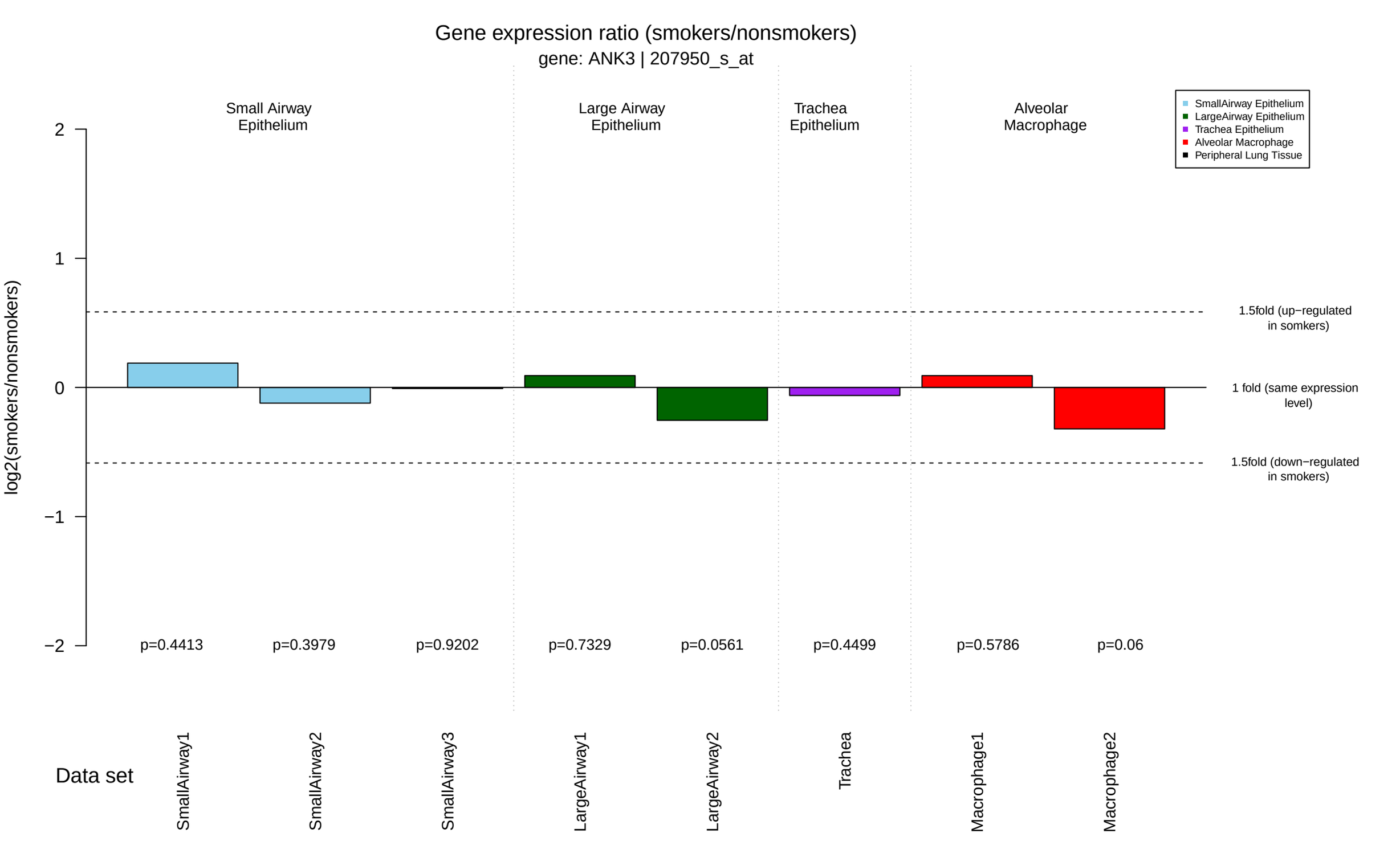
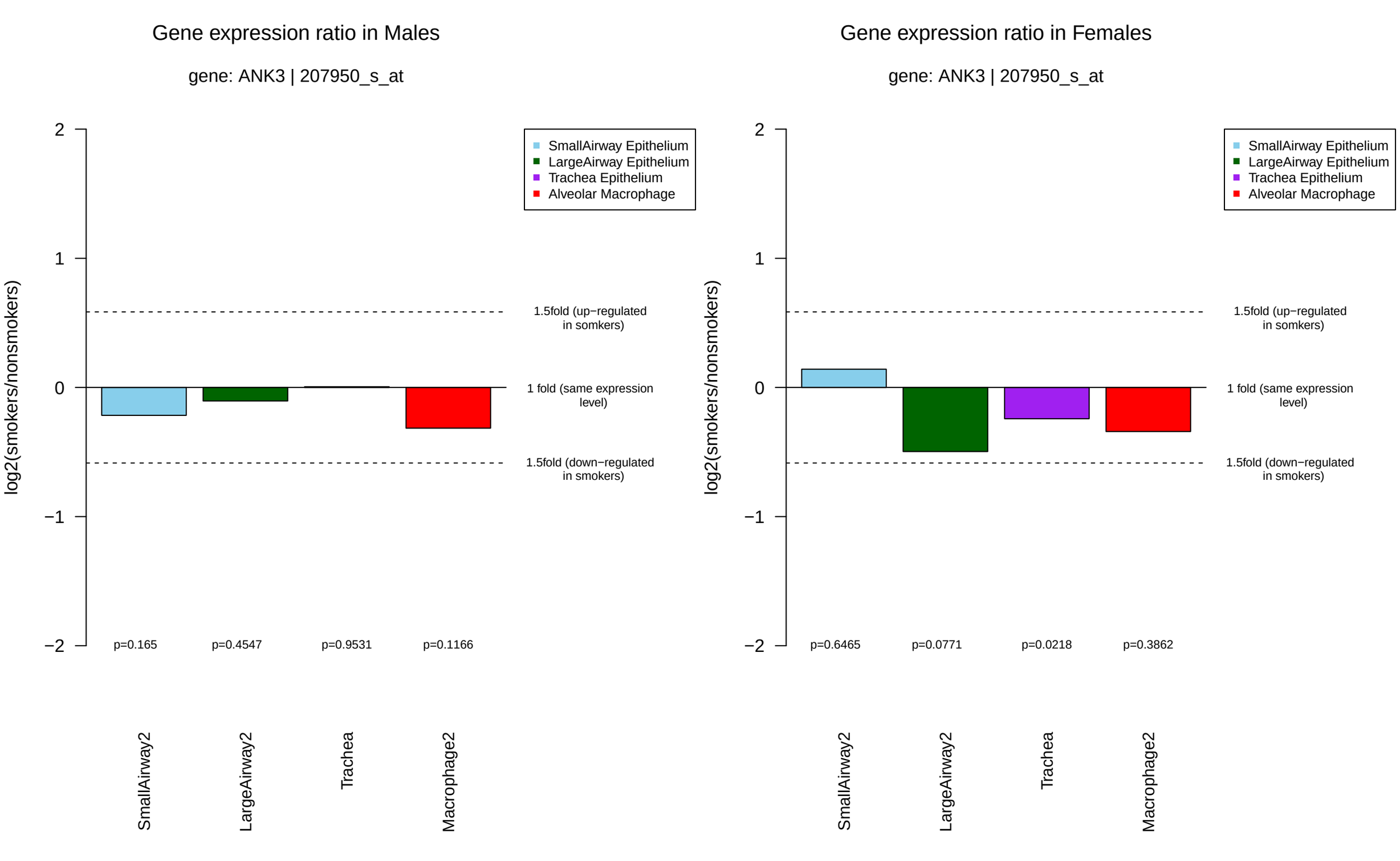
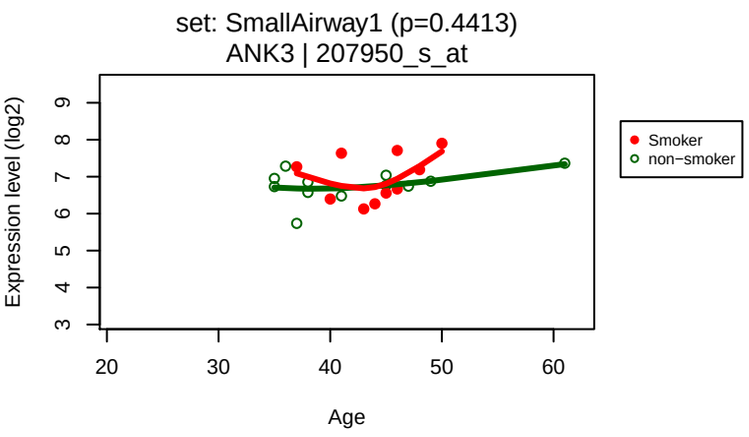
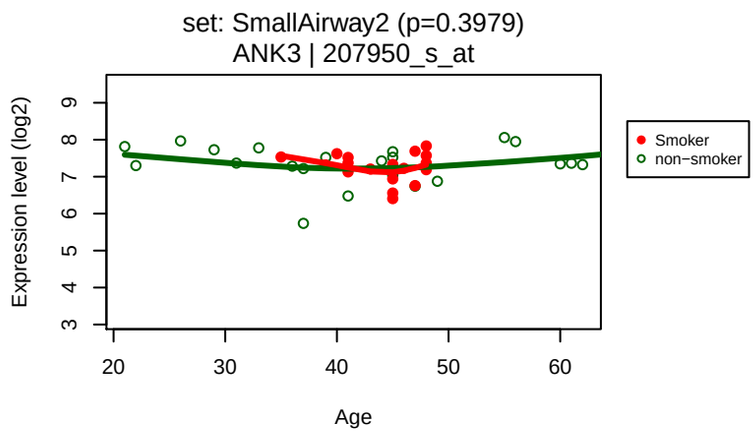
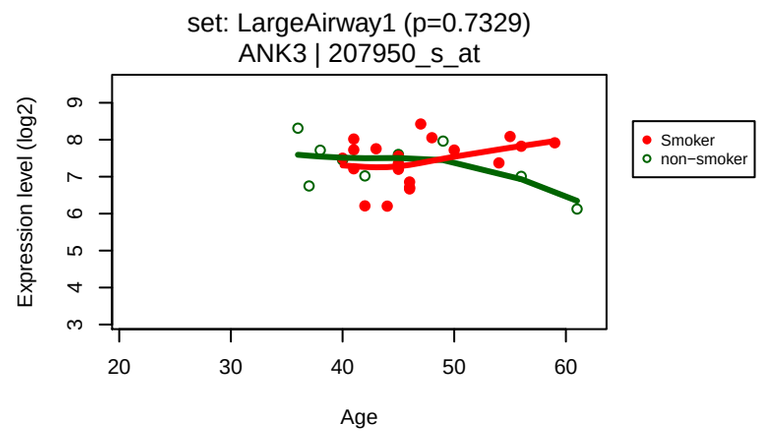

Gene expression of smokers and nonsmokers
gene: ANK3 | 207950_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4413
p=0.3979
p=0.9202
p=0.7329
p=0.0561
p=0.4499
p=0.5786
p=0.06
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

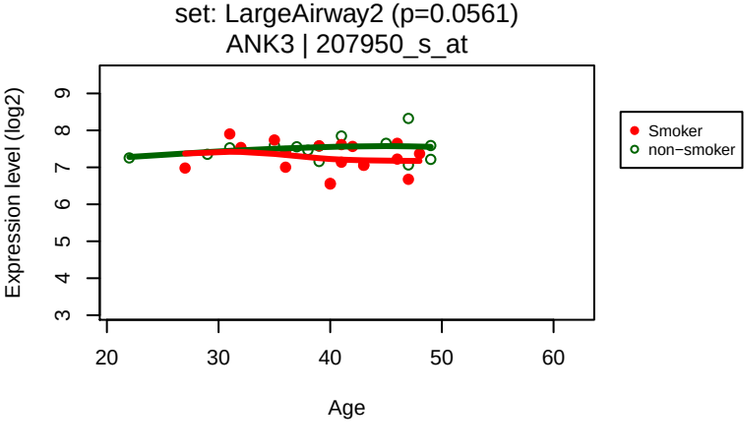

Gene expression ratio (smokers/nonsmokers)
gene: ANK3 | 207950_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4413
p=0.3979
p=0.9202
p=0.7329
p=0.0561
p=0.4499
p=0.5786
p=0.06
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

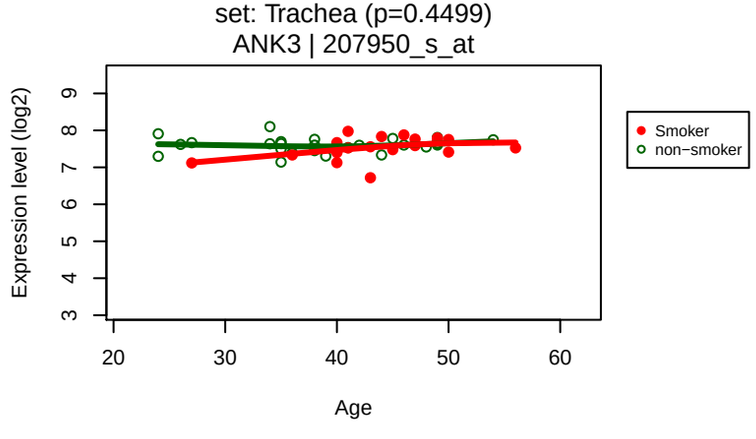

Gene expression ratio in Males
gene: ANK3 | 207950_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.165
p=0.4547
p=0.9531
p=0.1166
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ANK3 | 207950_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6465
p=0.0771
p=0.0218
p=0.3862
SmallAirway2
LargeAirway2
Trachea
Macrophage2

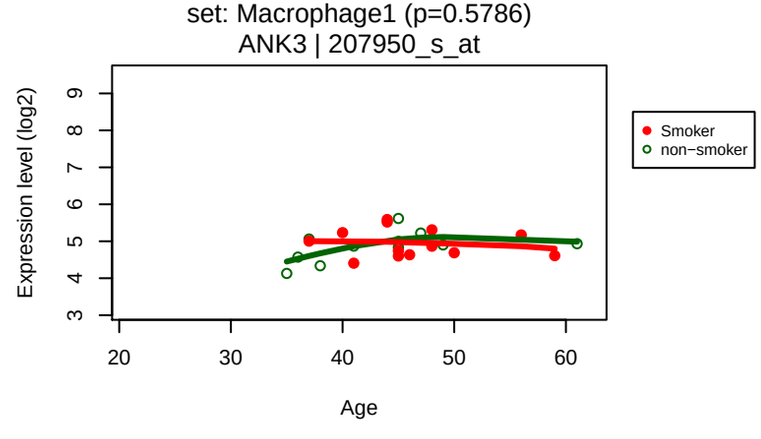

set: SmallAirway1 (p=0.4413)
ANK3 | 207950_s_at
Expression level (log2)
Age
Smoker
non−smoker

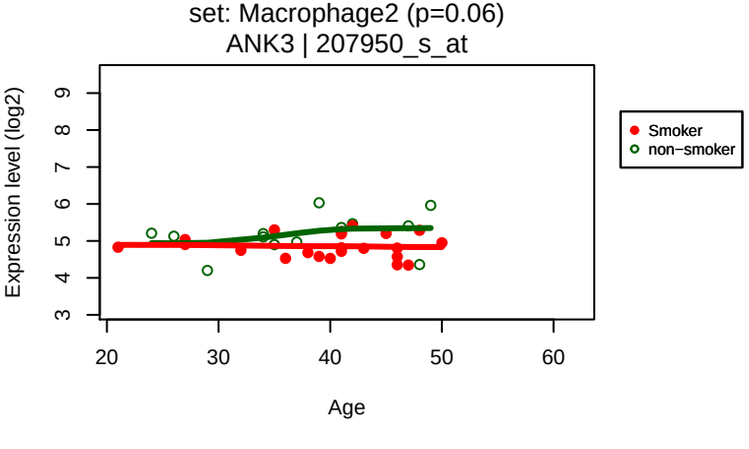

set: SmallAirway2 (p=0.3979)
ANK3 | 207950_s_at
Expression level (log2)
Age
Smoker
non−smoker

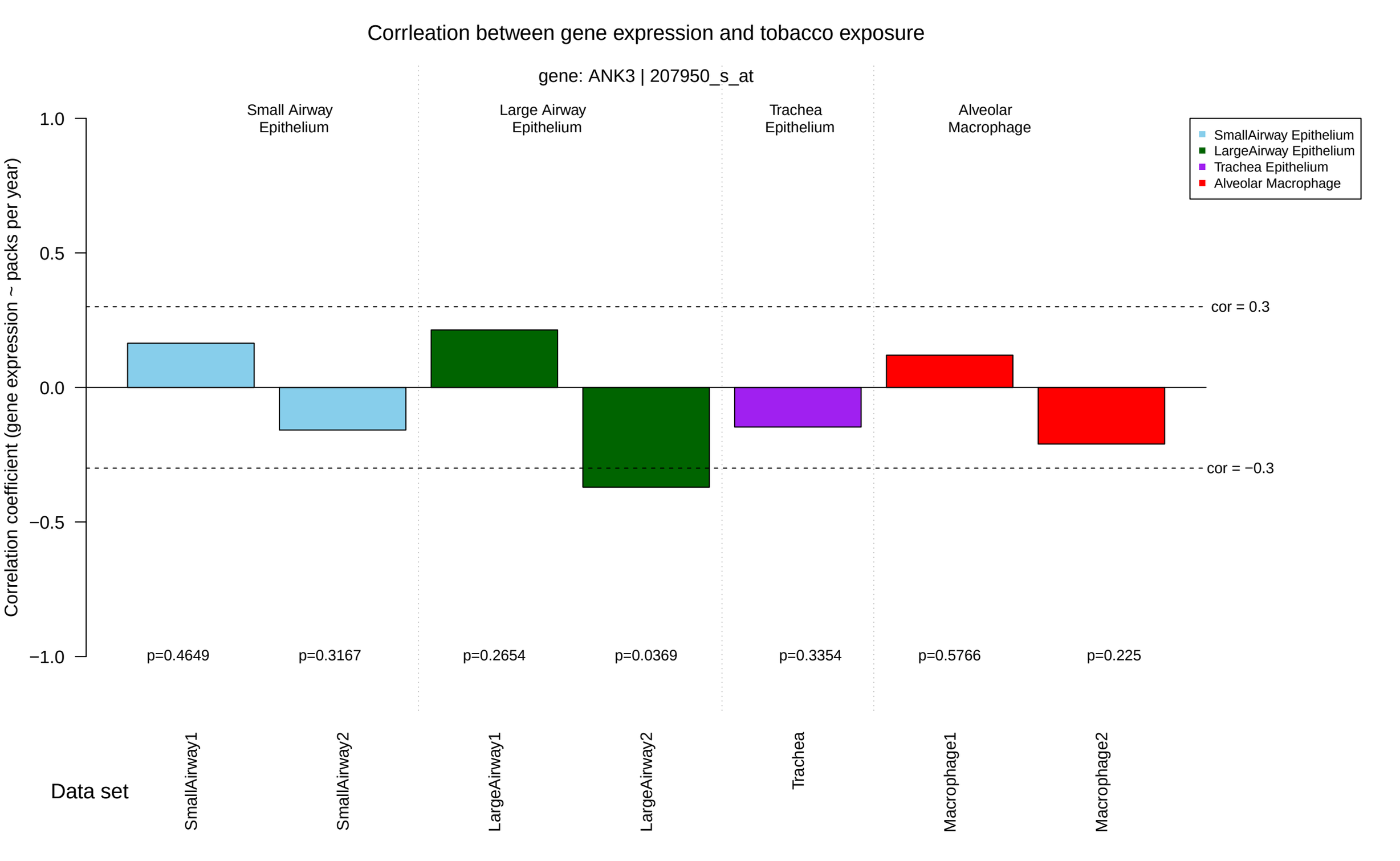

set: LargeAirway1 (p=0.7329)
ANK3 | 207950_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0561)
ANK3 | 207950_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4499)
ANK3 | 207950_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5786)
ANK3 | 207950_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.06)
ANK3 | 207950_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ANK3 | 207950_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4649
p=0.3167
p=0.2654
p=0.0369
p=0.3354
p=0.5766
p=0.225
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2